

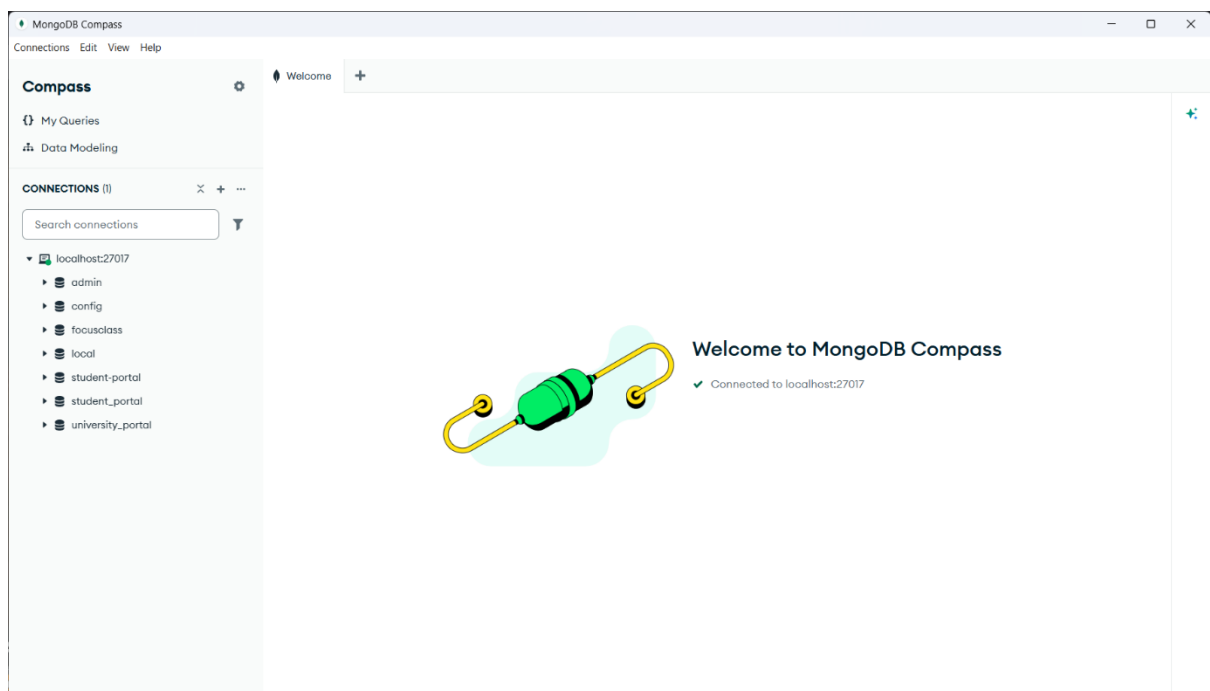
EMC WEEK 12 PROJECT BY VIGNESH S

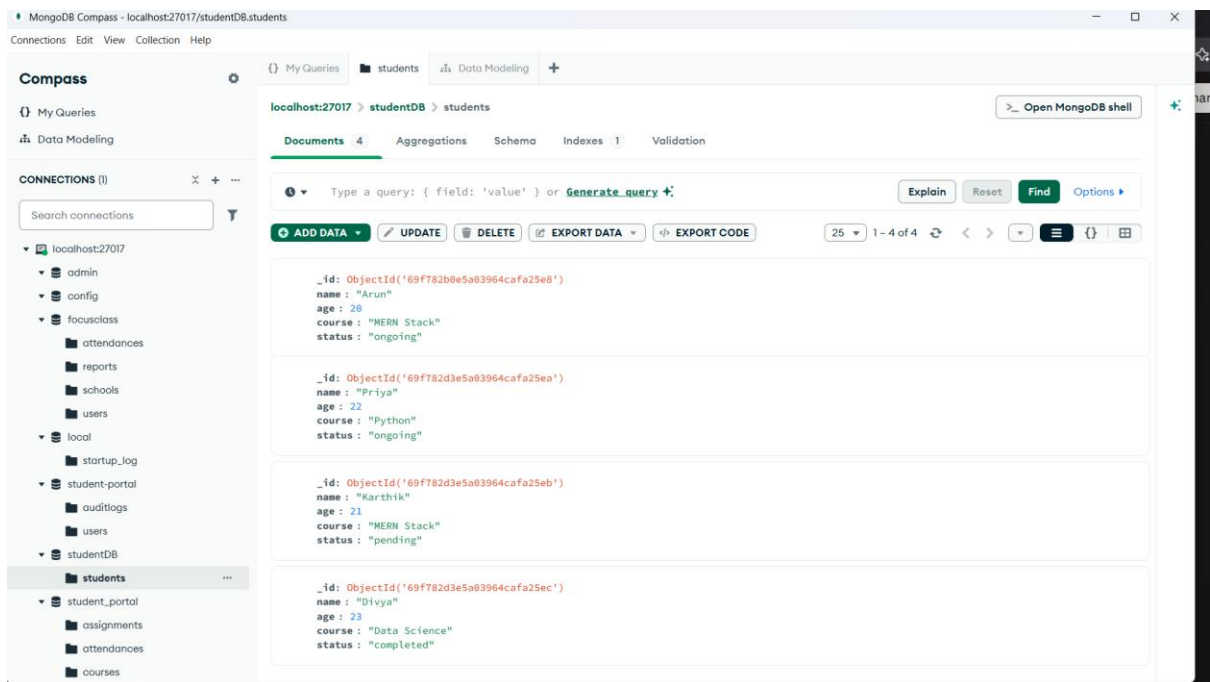
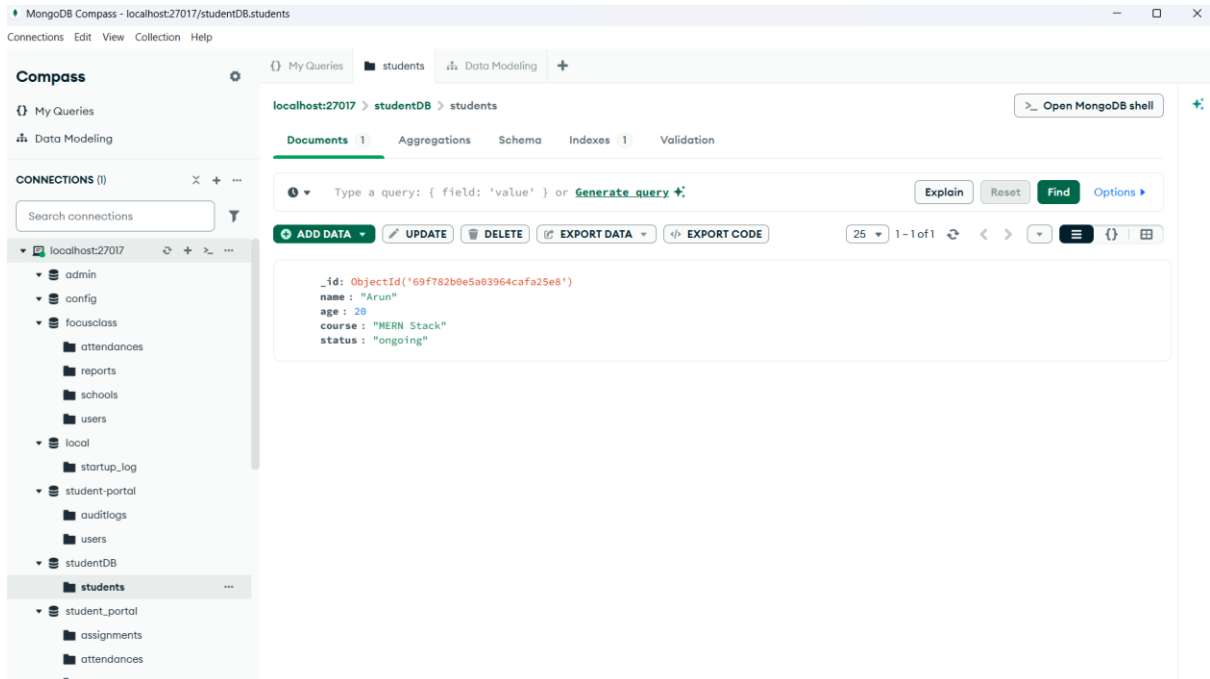
1. Database & Collection Setup:

- o Create a new database (e.g., studentDB)
- o Create a collection (e.g., students)

2. Insert Operations:

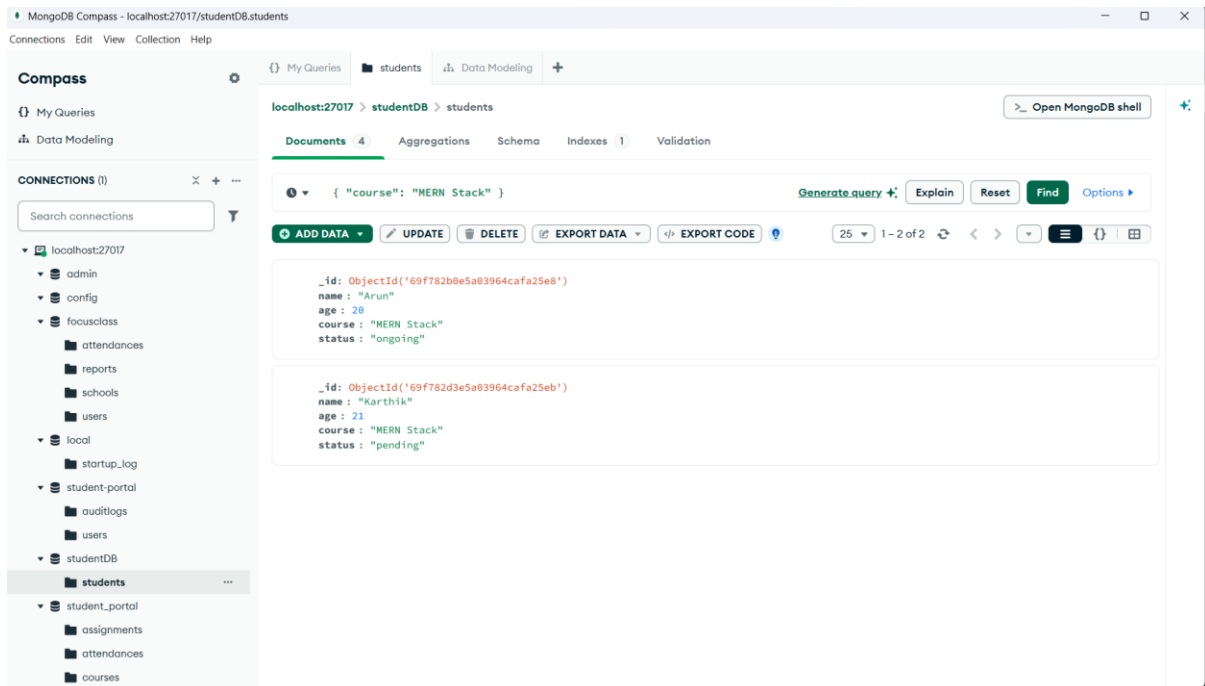
- o Insert one document into the students collection.
- o Insert multiple documents with fields like name, age, course, status.





3. Read Operations:

- o Fetch all documents from the collection.
- o Use filters to fetch specific data (e.g., students enrolled in “MERN Stack”).



4. Update Operations:

- o Update a single document (e.g., change the status of a student to completed).
- o Update multiple documents at once.

```
> db.students.updateOne(
  { name: "Hari" },
  { $set: { status: "completed" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.students.updateMany(
  { course: "MERN Stack" },
  { $set: { status: "completed" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 2,
  upsertedCount: 0
}
```

```
_id: ObjectId('69f782b0e5a03964cafa25e8')
name: "Arun"
age: 20
course: "MERN Stack"
status: "completed"
```

Task – 5

```
> db.students.deleteOne({ name: "David" })
< {
  acknowledged: true,
  deletedCount: 1
}
```

5(a). Delete one document where name = "David".

```
{ } My Queries students mongosh: localhost:27017 Data Modeling +
> _MONGOSH
> db.students.find()
< {
  _id: ObjectId('69f7868ab7384f8235efe514'),
  name: 'Bob',
  age: 24,
  course: 'Data Science',
  status: 'enrolled'
}
{
  _id: ObjectId('69f7868ab7384f8235efe515'),
  name: 'Charlie',
  age: 21,
  course: 'MERN Stack',
  status: 'completed',
  scholarship: true
}
{
  _id: ObjectId('69f7868ab7384f8235efe517'),
  name: 'Eve',
  age: 26,
  course: 'MERN Stack',
  status: 'inactive'
}
{
  _id: ObjectId('69f789a5a7252d9e1b1db482'),
  name: 'Arun',
  age: 20,
  course: 'MERN Stack',
  status: 'ongoing'
}
```

5(b). Delete all documents from the students collection .

```
}
> db.students.deleteMany({})
< {
  acknowledged: true,
  deletedCount: 4
}
```

```
  course: 'MERN Stack',
  status: 'ongoing'
}
db.students.deleteMany({})
< {
  acknowledged: true,
  deletedCount: 4
}
db.students.find()
db.students.find()
```

6. Query Operators: o Practice using \$gt, \$lt, \$in, \$and, \$or, \$exists in different queries.

```
}
> db.students.find({ age: { $gt: 21 } })
< {
  _id: ObjectId('69f78ba7a7252d9e1b1db484'),
  name: 'Bob',
  age: 24,
  course: 'Data Science',
  status: 'enrolled'
}
{
  _id: ObjectId('69f78ba7a7252d9e1b1db487'),
  name: 'Eve',
  age: 26,
  course: 'MERN Stack',
  status: 'inactive'
}
studentDB> |
```

```
> db.students.find({ age: { $lt: 22 } })
< {
  _id: ObjectId('69f78ba7a7252d9e1b1db483'),
  name: 'Arun',
  age: 20,
  course: 'MERN Stack',
  status: 'ongoing'
}
{
  _id: ObjectId('69f78ba7a7252d9e1b1db485'),
  name: 'Charlie',
  age: 21,
  course: 'MERN Stack',
  status: 'completed',
  scholarship: true
}
{
  _id: ObjectId('69f78ba7a7252d9e1b1db486'),
  name: 'David',
  age: 19,
  course: 'Python',
  status: 'enrolled'
}
studentDB> |
```

```
> db.students.find({ course: { $in: ["MERN Stack", "Python"] } })
< {
  _id: ObjectId('69f78ba7a7252d9e1bdb483'),
  name: 'Arun',
  age: 20,
  course: 'MERN Stack',
  status: 'ongoing'
}
{
  _id: ObjectId('69f78ba7a7252d9e1bdb485'),
  name: 'Charlie',
  age: 21,
  course: 'MERN Stack',
  status: 'completed',
  scholarship: true
}
{
  _id: ObjectId('69f78ba7a7252d9e1bdb486'),
  name: 'David',
  age: 19,
  course: 'Python',
  status: 'enrolled'
}
{
  _id: ObjectId('69f78ba7a7252d9e1bdb487'),
  name: 'Eve',
  age: 26
}
```

```
}
> db.students.find({
  $and: [
    { course: "MERN Stack" },
    { status: "completed" }
  ]
})
< {
  _id: ObjectId('69f78ba7a7252d9e1bdb485'),
  name: 'Charlie',
  age: 21,
  course: 'MERN Stack',
  status: 'completed',
  scholarship: true
}
studentDB>
```

```

> db.students.find({
  $or: [
    { course: "MERN Stack" },
    { scholarship: true }
  ]
})
< {
  _id: ObjectId('69f78ba7a7252d9e1b1db483'),
  name: 'Arun',
  age: 20,
  course: 'MERN Stack',
  status: 'ongoing'
}
{
  _id: ObjectId('69f78ba7a7252d9e1b1db485'),
  name: 'Charlie',
  age: 21,
  course: 'MERN Stack',
  status: 'completed',
  scholarship: true
}
{
  _id: ObjectId('69f78ba7a7252d9e1b1db487'),
  name: 'Eve',
  age: 26,
  course: 'MERN Stack',
  status: 'inactive'
}

```

```

> db.students.find({ scholarship: { $exists: true } })
< {
  _id: ObjectId('69f78ba7a7252d9e1b1db485'),
  name: 'Charlie',
  age: 21,
  course: 'MERN Stack',
  status: 'completed',
  scholarship: true
}
studentDB>

```

Task 7 - Use Case:

o Create a small use case like a Library System or E-Commerce product listing and apply insert, update, and search operations.

```

}
> use libraryDB
db.createCollection("books")
< switched to db libraryDB
libraryDB>

```

```

>_MONGOSH
> use libraryDB
db.createCollection("books")
< switched to db libraryDB
> db.books.insertMany([
  { title: "MongoDB Basics",    author: "John Doe",   year: 2023, genre: "Technology", available: true },
  { title: "Node.js in Action", author: "Jane Smith", year: 2021, genre: "Technology", available: true },
  { title: "Harry Potter",      author: "J.K. Rowling", year: 1997, genre: "Fantasy",   available: false },
  { title: "Clean Code",        author: "Robert Martin", year: 2008, genre: "Technology", available: true }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f78d95a7252d9e1b1db488'),
    '1': ObjectId('69f78d95a7252d9e1b1db489'),
    '2': ObjectId('69f78d95a7252d9e1b1db48a'),
    '3': ObjectId('69f78d95a7252d9e1b1db48b')
  }
}
> db.books.find()
< {
  _id: ObjectId('69f78d95a7252d9e1b1db488'),
  title: 'MongoDB Basics',
  author: 'John Doe',
  year: 2023,
  genre: 'Technology',
  available: true
}
{
  _id: ObjectId('69f78d95a7252d9e1b1db489'),

```

```

>_MONGOSH
  year: 2023,
  genre: 'Technology',
  available: true
}
{
  _id: ObjectId('69f78d95a7252d9e1b1db489'),
  title: 'Node.js in Action',
  author: 'Jane Smith',
  year: 2021,
  genre: 'Technology',
  available: true
}
{
  _id: ObjectId('69f78d95a7252d9e1b1db48a'),
  title: 'Harry Potter',
  author: 'J.K. Rowling',
  year: 1997,
  genre: 'Fantasy',
  available: false
}
{
  _id: ObjectId('69f78d95a7252d9e1b1db48b'),
  title: 'Clean Code',
  author: 'Robert Martin',
  year: 2008,
  genre: 'Technology',
  available: true
}
libraryDB>

```

```
{ } My Queries  students  mongosh: localhost:27017  Data Modeling  +

>_MONGOSH

  '3': ObjectId('69f78d95a7252d9e1b1db48b')
}
}
> db.books.find()
< {
  _id: ObjectId('69f78d95a7252d9e1b1db488'),
  title: 'MongoDB Basics',
  author: 'John Doe',
  year: 2023,
  genre: 'Technology',
  available: true
}
{
  _id: ObjectId('69f78d95a7252d9e1b1db489'),
  title: 'Node.js in Action',
  author: 'Jane Smith',
  year: 2021,
  genre: 'Technology',
  available: true
}
{
  _id: ObjectId('69f78d95a7252d9e1b1db48a'),
  title: 'Harry Potter',
  author: 'J.K. Rowling',
  year: 1997,
  genre: 'Fantasy',
  available: false
}
{
  _id: ObjectId('69f78d95a7252d9e1b1db48b'),
  title: 'Clean Code',
  author: 'Robert C. Martin',
  year: 2019,
  genre: 'Technology',
  available: true
}
```

```
>_MONGOSH

}
> db.books.find({ available: true })
< {
  _id: ObjectId('69f78d95a7252d9e1b1db488'),
  title: 'MongoDB Basics',
  author: 'John Doe',
  year: 2023,
  genre: 'Technology',
  available: true
}
{
  _id: ObjectId('69f78d95a7252d9e1b1db489'),
  title: 'Node.js in Action',
  author: 'Jane Smith',
  year: 2021,
  genre: 'Technology',
  available: true
}
{
  _id: ObjectId('69f78d95a7252d9e1b1db48a'),
  title: 'Harry Potter',
  author: 'J.K. Rowling',
  year: 1997,
  genre: 'Fantasy',
  available: true
}
{
  _id: ObjectId('69f78d95a7252d9e1b1db48b'),
  title: 'Clean Code',
  author: 'Robert C. Martin',
  year: 2019,
  genre: 'Technology',
  available: true
}
```

>_MONGOSH

1

```
    genre: 'Technology',
    available: true
  }
> db.books.find({ genre: "Technology" })
< {
  _id: ObjectId('69f78d95a7252d9e1bdb488'),
  title: 'MongoDB Basics',
  author: 'John Doe',
  year: 2023,
  genre: 'Technology',
  available: true
}
{
  _id: ObjectId('69f78d95a7252d9e1bdb489'),
  title: 'Node.js in Action',
  author: 'Jane Smith',
  year: 2021,
  genre: 'Technology',
  available: true
}
{
  _id: ObjectId('69f78d95a7252d9e1bdb48b'),
  title: 'Clean Code',
  author: 'Robert Martin',
  year: 2008,
  genre: 'Technology',
  available: true
}
libraryDB>
```